Supplementary Material

From Feedback to Revision: When Multi-Agent Roles Become Load-Bearing

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WHAT THIS DOCUMENT IS

The main paper states every claim it makes and is complete without this file. What it cannot carry, at eight pages, is the full numeric backing behind five of its sentences. That backing is here, one table per sentence, each reproducing the pointer the main text gives.

Every table below is **generated** from one hash-manifested frozen snapshot by a single script; no number in either document is hand-typed. Each table carries its own provenance header in the source. The tables are reproduced here unchanged — same files, same bytes — so a number cannot drift between the paper and its supplement.

Reading the denominators. Two populations recur and are not interchangeable. The *marginal* population is every trajectory in a cell, with each condition on its own denominator. The *paired* population is only those problems first-pass wrong under *both* conditions, on one shared denominator; that is the actual test, and it is what every reported p-value is computed on (exact McNemar on discordant pairs). Where a table shows both, they are labeled and must not be read across. Conflating them has already produced one error in this project.

Recovery denominator. Throughout, recovery is conditioned on a recorded *first* evaluator verdict of incorrect — including trajectories that later died in regeneration, which score as failures under the pre-registered rule. Excluding those was a real defect in an earlier analysis and it moved results materially; it is noted here so a reader recomputing from the released data gets the same denominator.

1 THE PRE-REGISTERED ENDPOINTS IN FULL

The main paper reports, in its Intervention section, that the pre-registered primary endpoint P4 — the actor × intervention interaction on accuracy — is null in all three pairwise difference-in-differences, and that the P1a manipulation objective failed. Table 1 is the complete endpoint table promised there: all five endpoints as written before any intervention data existed, what each one returned, and the Holm-corrected verdict.

Two things in it deserve to be read rather than skimmed. First, only one of the three P4 contrasts is an *informative* null: the two involving Llama-3.3 cannot detect an interaction of any plausible size at their sample size, and are uninformative rather than evidence of absence. Second, the contrast this paper is organized around — role-structured versus role-free — was pre-registered as **secondary and exploratory**. It is labeled post-hoc in the main paper and is labeled post-hoc here.

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Table 1: Every pre-registered endpoint, and what became of it. Fixed in writing before any intervention data existed. P4 — the pre-registered primary test — is null, before and after the pre-registered Holm correction, and we report it as the failure of the thesis it was written to test. P1 failed *as written* and was re-designated rather than dropped: the interface forbids the planner emitting an *answer*, and nothing in it stops the planner *computing* — a dissociation we report as a finding. P2 and P5 are reported as effects and associations and carry no p-value, so the correctable family is P3's three contrasts and P4's three difference-in-differences. The MDE column is a detectability scale on a different population from the effect column — it is computed on problems shared by both actors, while each difference-in-difference uses each actor's own problem set; neither can be derived from the other.

endpoint	disposition
P1 intervention → fidelity (also the manipulation check)	**failed as written; re-designated.** The interface forbids an *answer* field, not computation. Re-scoped to answer authorship, which it drives to 0.000 in all three actors (Table 4). Reported as a second dissociation, not as a pass.
P2 actor × interv. on fidelity	not estimable across the panel: the fidelity clauses are condition-specific, so no common scale exists. Reported as unresolved rather than substituted.

P3 intervention → accuracy (enforced − role-structured), Holm across three:

	raw p	Holm p	verdict
gemma-4-31B	0.004	0.012	reject
gpt-oss-120b	0.582	0.789	retain
Llama-3.3-70B	0.394	0.789	retain

P4 actor × intervention on accuracy — *the pre-registered primary*, Holm across three:

	difference-in-diff. [CI]	raw p	Holm p	$MDE_{80\%}$	null is
gpt-oss-120b − gemma-4-31B	+0.0175 [-0.0050,+0.0406]	0.145	0.29	0.0325	informative
Llama-3.3-70B − gemma-4-31B	+0.0400 [+0.0006,+0.0800]	0.0553	0.166	0.0693	uninformative
Llama-3.3-70B − gpt-oss-120b	+0.0225 [-0.0181,+0.0631]	0.286	0.29	0.0748	uninformative

P5 Δfidelity vs Δoutcome	association only, pre-declared as non-causal: both sides are outcomes of the same intervention. No p-value is reported and none is corrected.

Only one of the three P4 nulls is informative. The largest per-actor main effect is small enough that a true interaction is plausibly ≤ 0.040; only the gpt-oss/gemma-4 contrast can see an effect that size. The two Llama-3.3 contrasts cannot detect an interaction of any plausible magnitude and are uninformative rather than evidence of absence.

2 FIRST PASS AND RECOVERY, BOTH POPULATIONS

Table 2 gives the per-actor numbers behind the main paper's central result, in both populations, so that the marginal rates and the paired test can be checked against each other.

The gemma-4 row is the one the paper's headline rests on. Its first-pass rates are equal to four decimal places, which is why the recovery contrast in that actor is an exact reparameterization of the randomized intent-to-treat effect rather than a conditional on a post-treatment variable: with equal first-pass rates the two populations coincide in expectation, and the role-free team does not enter recovery already behind. That argument does *not* transfer to the other two actors, and the paper does not make it there.

Table 2: First-pass accuracy and recovery, role-structured versus role-free. The two panels have different denominators and must not be read across. *Upper:* the marginal population — every trajectory in the cell, first pass over those receiving an evaluator verdict, recovery over those whose *first* verdict was incorrect (n_{fpw} in parentheses, computed per condition and therefore *not* a shared denominator). *Lower:* the paired population — only problems first-pass wrong under *both* conditions, one shared denominator n_{both}, which is the actual test. The recovery denominator counts every trajectory with a recorded incorrect first verdict, including those that later died in regeneration; these score 0 under the pre-registered rule. Pre-registered scoring throughout; p is exact McNemar on discordant pairs.

Marginal population — each condition on its own denominator

	first pass			recovery | first verdict wrong		
actor	role-struct.	role-free	difference	role-struct. (n)	role-free (n)	difference
gemma-4-31B	0.802	0.802	+0.0000 [−0.0156, +0.0150]	0.222 (316)	0.051 (316)	−0.1709 [−0.2247, −0.1171]
gpt-oss-120b	0.713	0.763	+0.0503 [+0.0339, +0.0666]	0.324 (463)	0.247 (380)	−0.0766 [−0.1361, −0.0153]
Llama-3.3-70B	0.308	0.316	+0.0077 [−0.0283, +0.0437]	0.014 (276)	0.015 (268)	+0.0004 [−0.0214, +0.0192]

Paired population — problems first-pass wrong under **both** conditions; one shared denominator

actor	n_{both}	role-struct.	role-free	role-struct. only	role-free only	exact p
gemma-4-31B	237	0.160	0.017	37	3	1.95×10^{-8}
gpt-oss-120b	327	0.242	0.202	52	39	0.208
Llama-3.3-70B	245	0.004	0.016	1	4	0.375

Recovery conditions on a post-treatment variable and is descriptive, never causal. No cell recovers before its first oracle verdict.

3 CROSS-DOMAIN REPLICATION

Table 3 is the SciBench replication in full. The main paper carries gemma-4 in Figure 2 and in prose; this is both halves with their denominators.

The domain contract is *not* byte-identical to Omni-MATH. SciBench problems are free-response college science with units, so the answer normalizer and the problem framing are domain-adapted; the role contracts, topology, budget, decoding and evaluator protocol are unchanged. This is a replication of the *mechanism* under a matched protocol, not a rerun of identical prompts, and the main text says so.

Table 3: The dissociation replicates out of domain. SciBench, the complete 574-problem matched slice of free-response college science; two actors, 2296 trajectories, frozen separately from the Omni-MATH snapshot. The two panels have different populations. First pass is paired over all 574 problems; paired recovery is over only those first-pass wrong under *both* conditions (n_{both}), which is why its n is roughly a sixth the size. Pre-registered scoring; p is exact McNemar on discordant pairs. gpt-oss's recovery cell cannot resolve its own effect at the realized n_{both} — its point estimate is in the same direction and nearly the same size as Omni-MATH's, so it is underpowered rather than a non-replication. Llama-3.3 was not run in this domain.

	first pass (paired, all problems)					recovery (paired on first-pass wrong under both)						
actor	n	role-str.	role-free	diff.	exact p	n_{both}	role-str.	role-free	diff.	r-s. only	r-f. only	exact p
gemma-4-31B	574	0.794	0.794	+0.0000	1	99	0.212	0.020	−0.1919	21	2	6.60×10^{-5}
gpt-oss-120b	574	0.784	0.789	+0.0052	0.784	96	0.219	0.135	−0.0833	17	9	0.169

4 STRUCTURAL ENFORCEMENT: BOTH SCORINGS, AND THE ATTRITION THAT SEPARATES THEM

The main paper reports that enforcement un-freezes the answer in all three actors but does not become end-to-end accuracy, and that the apparent accuracy benefit is survivorship. Table 4 is the evidence for both halves: the manipulation check, the one-sided attrition, and the same accuracy column under both scorings with intervals.

This is the table behind the sentence in the main text that the enforced recovery and first-pass rates **do not compose** into the accuracy column. They cannot: trajectories that never finish are never graded, so the three quantities do not share a denominator and the $(1 - \text{first-pass}) \times \text{recovery}$ identity does not hold. The pre-registration fixed in advance that unparseable trajectories count as incorrect rather than being dropped, and that rule is what makes the as-graded reading survivorship rather than a finding.

Table 4: Structural enforcement. Enforcing the division of labor at the interface. *Manipulation check:* the rate at which the planner authors an answer, under enforcement and (in parentheses) when roles are merely deleted — enforcement drives it to zero in all three actors, so the intervention engaged. *Frozen answer:* the team never changed its answer string, among first-pass-wrong trajectories (role-free/role-structured in parentheses). *Recovery:* role-structured in parentheses, on the denominator of Table 2. The two accuracy columns are different populations, and the attrition column is why they disagree. The pre-registered rule scores never-graded trajectories incorrect; the as-graded column drops them and is a *sensitivity analysis only* — its gains are conditional on parsing successfully, and enforcement is the only condition where that removes a material fraction of the data. Enforcement restores recovery for Llama-3.3; for the other two actors the direction is consistent but not resolvable against that attrition.

	planner authors	frozen answer	recovery		final accuracy, enforced − role-structured	
actor	answer (role-free) manipulation check	(r-f./r-s.) first-pass wrong	(role-str.) first-pass wrong	attrition	pre-registered (primary)	as-graded (sensitivity)
gemma-4-31B	0.000 (0.986)	0.119 (0.845/0.288)	0.314 (0.222)	8.0%	−0.0225 [-0.0387,-0.0063]	+0.0190 [+0.0054,+0.0333]
gpt-oss-120b	0.000 (0.916)	0.092 (0.208/0.052)	0.395 (0.324)	9.9%	−0.0050 [-0.0219,+0.0112]	+0.0361 [+0.0215,+0.0507]
Llama-3.3-70B	0.000 (0.958)	0.072 (0.806/0.750)	0.139 (0.014)	17.0%	+0.0175 [-0.0200,+0.0550]	+0.0453 [+0.0030,+0.0876]

Attrition elsewhere is negligible: 0.0–0.7% with roles, 0.0–2.5% with roles removed. The as-graded column turns every sign positive; the pre-registration fixed in advance that unparseable trajectories count incorrect, and **the benefit appears only after conditioning on successful parsing.**

Table 5: Against a budget-matched single agent. Both directions, which must be read together. A collapsed gpt-oss team beats its own single agent, so three agents emitting one answer is not functionally one agent; but gemma-4's team is worse than its own single agent in both conditions — and gemma-4 carries the recovery result. Provenance: this panel is not from the frozen snapshot; it comes from a previously released trace corpus, joined by problem. Its two actor arms were built from different source commits, so each direction is a sound directional guard while *the gap between actors is not* — no statement about relative magnitude across actors is supported here. The comparison is conservative on the decisive axis: the single agent draws *more* oracle feedback and about half the actor calls.

organization	team	single agent	difference
gemma-4-31B			
role-structured	0.8462	0.8688	−0.0225
	$[-0.0400, -0.0056]$, $p = 0.0122$, $n = 1600$		
role-free	0.8125	0.8688	−0.0563
	$[-0.0737, -0.0394]$, $p = 1.22 \times 10^{-10}$, $n = 1600$		
gpt-oss-120b			
role-structured	0.8031	0.7900	+0.0131
	$[-0.0044, +0.0300]$, $p = 0.152$, $n = 1600$		
role-free	0.8194	0.7900	+0.0294
	$[+0.0125, +0.0456]$, $p = 0.000678$, $n = 1600$		

Realized call budget

actor	protocol	evaluator	actor
gemma-4-31B	single agent	1.66	3.16
gemma-4-31B	role-structured team	1.11	6.76
gemma-4-31B	role-free team	1.95	5.87
gpt-oss-120b	single agent	1.69	3.40
gpt-oss-120b	role-structured team	1.35	7.54
gpt-oss-120b	role-free team	1.85	6.25

5 THE SINGLE-AGENT COMPARATOR

Table 5 is the budget-matched single-agent reference. The main paper uses it in one direction only — as a guard against reading "collapse costs nothing" as "the system was one agent all along" — and this table is why that restraint is necessary.

The comparator comes from a **previously released corpus whose two actor arms were built from different source commits**. Each direction within an actor is therefore a sound guard, and *the gap between actors is not.* The result cuts both ways: a collapsed gpt-oss team beats its own single agent, while gemma-4's is significantly worse in both conditions. Neither "multi-agent systems are just single agents" nor "more agents help" survives contact with it.

Table 6: Recovery by pooled accumulated-context quartile (supplement; the panel behind the context caveat in R2). Denominator is the first-pass-wrong population. Context is the total character count of all planner, executor and reviewer messages in the trajectory; quartile cut points are computed *once per actor over both conditions pooled*. This is directional only. Accumulated context is nearly a function of condition — the role-free team talks less because it stops reworking — so the strata are not balanced and this comparison cannot separate context from condition. It is reported because it is the obvious competing explanation and because it does not behave uniformly: the gap survives in the lower quartiles, but in gpt-oss's top quartile the ordering reverses.

	role-structured		role-free		
context quartile	recovery	*n*	recovery	*n*	difference
gemma-4-31B					
Q1 (least context)	0.662	65	0.151	93	+0.511
Q2	0.280	50	0.009	108	+0.271
Q3	0.047	85	0.014	73	+0.033
Q4 (most)	0.078	116	0.000	42	+0.078
gpt-oss-120b					
Q1 (least context)	0.923	13	0.258	198	+0.665
Q2	0.656	128	0.293	82	+0.363
Q3	0.262	164	0.191	47	+0.071
Q4 (most)	0.070	158	0.189	53	−0.119
Llama-3.3-70B					
Q1 (least context)	0.100	40	0.031	96	+0.069
Q2	0.000	86	0.000	50	+0.000
Q3	0.000	103	0.030	33	−0.030
Q4 (most)	0.000	47	0.000	89	+0.000

6 ACCUMULATED CONTEXT AS A COMPETING EXPLANATION

Accumulated context is the obvious competing cause of recovery failure, and Table 6 is the full panel promised in the main text.

It is reported with a caveat that is not decoration. Quartile cut points are computed over both conditions *pooled*, and accumulated context is nearly a function of condition — the role-free team accumulates less because it stops reworking. The strata are therefore not balanced, and this comparison cannot separate context from condition. It is **directional only**, and it is included because it does not behave uniformly: the gap survives in the lower quartiles, but in gpt-oss's top quartile the ordering reverses. A stratification that had come out cleanly everywhere would have been weaker evidence, not stronger, because it could not have failed.

7 AI ASSISTANCE

This work used LLM agents substantially in hypothesis formation, experimental design, execution, analysis and writing, under human direction and approval. The full disclosure — tools and versions as retained, what the agents did at each stage, what the human author directed and approved, which prompts are released and which were not retained, and the instrumentation defects this process found in its own work — accompanies this submission as a separate anonymous document.